

## Exercise 2.7

Show

$$[H, a_p^\dagger] = \omega_p a_p^\dagger, \quad [H, a_p] = -\omega_p a_p$$

The Hamiltonian is 
$$H = \int \frac{d^3p}{(2\pi)^3} \omega_p \left[ a_p^\dagger a_p + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right]$$

We choose to ignore the delta-function term

$$\Rightarrow H = \int d^3p / (2\pi)^3 \omega_p [a_p^\dagger a_p]$$

$$\begin{aligned} \Rightarrow [H, a_p^\dagger] &= \int d^3q / (2\pi)^3 \omega_q (a_q^\dagger a_q a_p^\dagger - a_p^\dagger a_q^\dagger a_q) \quad \parallel \text{We use } [a_p^\dagger, a_q^\dagger] = 0 \\ &= \int d^3q / (2\pi)^3 \omega_q a_q^\dagger [a_q, a_p^\dagger] \quad \parallel [a_q, a_p^\dagger] = (2\pi)^3 \delta^{(3)}(q-p) \end{aligned}$$

$$[H, a_p^\dagger] = \int d^3q \omega_q a_q^\dagger \delta^{(3)}(q-p) = \underline{\underline{\omega_p a_p^\dagger}} \quad \square$$

$$[H, a_p] = \int d^3q / (2\pi)^3 \omega_q (a_q^\dagger a_q a_p - a_p a_q^\dagger a_q) \quad \parallel [a_q, a_p] = 0$$

$$\begin{aligned} &= \int d^3q / (2\pi)^3 \omega_q \underbrace{(a_q^\dagger a_p - a_p a_q^\dagger)}_{-[a_p, a_q^\dagger]} a_q \\ &\quad - [a_p, a_q^\dagger] = -(2\pi)^3 \delta^{(3)}(p-q) \end{aligned}$$

$$= - \int d^3q \omega_q a_q \delta^{(3)}(p-q) = -\omega_p a_p$$

Show  $|p\rangle = a_p^\dagger |0\rangle$  has energy  $\omega_p = \sqrt{p^2 + m^2}$ 

Energy is eigenvalue of Hamiltonian:

$$\hat{H}|p\rangle = E|p\rangle \Rightarrow [\hat{H}, a_p^\dagger]|0\rangle = \omega_p a_p^\dagger |0\rangle = \hat{H} a_p^\dagger |0\rangle - \underbrace{a_p^\dagger \hat{H}}_{=0} |0\rangle$$

H annihilates the vacuum state since  $a_p|0\rangle = 0$

So we get

$$\omega_p a_p^\dagger |0\rangle = \hat{H} a_p^\dagger |0\rangle \quad || a_p^\dagger |0\rangle = |p\rangle$$

$$\omega_p |p\rangle = \hat{H} |p\rangle$$

Finally show that  $\hat{P} |p\rangle = p |p\rangle$

$$\hat{P} = + \int \frac{d^3p}{(2\pi)^3} p a_p^\dagger a_p$$

$$\hat{P} |p\rangle = \hat{P} a_p^\dagger |0\rangle = \left[ \int \frac{d^3q}{(2\pi)^3} q a_q^\dagger a_q a_p^\dagger \right] |0\rangle \quad || a_q a_p^\dagger = [a_q, a_p^\dagger] + a_p^\dagger a_q$$

$$= \int \frac{d^3q}{(2\pi)^3} q a_q^\dagger ([a_q, a_p^\dagger] + a_p^\dagger a_q) |0\rangle \quad || [a_q, a_p^\dagger] = (2\pi)^3 \delta^{(3)}(q-p)$$

this annihilates the vacuum state:  $a_q |0\rangle = 0$

$$= \left[ \int \frac{d^3q}{(2\pi)^3} q a_q^\dagger (2\pi)^3 \delta^{(3)}(q-p) \right] |0\rangle$$

$$\hat{P} |p\rangle = p a_p^\dagger |0\rangle = \underline{\underline{p |p\rangle}} \quad \square$$

### Exercise 2.8

Let  $\phi$  be a scalar field satisfying the field equation

$$(\partial^\mu \partial_\mu + \mu^2) \phi(x) = 0$$

Show

$$\Delta_F(x-y) = \langle 0 | T(\phi(x) \phi(y)) | 0 \rangle \equiv \theta(x_0 - y_0) \langle 0 | \phi(x) \phi(y) | 0 \rangle + \theta(y_0 - x_0) \langle 0 | \phi(y) \phi(x) | 0 \rangle$$

Can be written as

$$\Delta_F(x-y) = \int \frac{d^4k}{(2\pi)^4} e^{-ik(x-y)} \frac{i}{k^2 - \mu^2 + i\epsilon}$$

$$\phi(\vec{x}, t) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left( \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_k}} (a_{\vec{k}} e^{-i\vec{k}\cdot\vec{x}} + a_{\vec{k}}^\dagger e^{i\vec{k}\cdot\vec{x}}) \right)$$

$$\langle 0 | \phi(\vec{x}, t) \phi(\vec{y}, t) | 0 \rangle = \langle 0 | \left[ \int \frac{d^3k}{(2\pi)^3} \frac{d^3q}{2\sqrt{E_k E_q}} (a_{\vec{k}} e^{-i\vec{k}\cdot\vec{x}} + a_{\vec{k}}^\dagger e^{i\vec{k}\cdot\vec{x}}) (a_{\vec{q}} e^{-i\vec{q}\cdot\vec{y}} + a_{\vec{q}}^\dagger e^{i\vec{q}\cdot\vec{y}}) \right] | 0 \rangle$$

$\langle 0 | a_{\vec{k}}^\dagger = 0$  &  $a_{\vec{q}} | 0 \rangle = 0$  which means only the  $a_{\vec{k}} a_{\vec{q}}^\dagger e^{-i\vec{k}\cdot\vec{x} + i\vec{q}\cdot\vec{y}}$  term survives inside the integral. We also note that

$$a_{\vec{k}} a_{\vec{q}}^\dagger = [a_{\vec{k}}, a_{\vec{q}}^\dagger] + a_{\vec{q}}^\dagger a_{\vec{k}} \quad \& \quad \langle 0 | a_{\vec{q}}^\dagger a_{\vec{k}} | 0 \rangle = 0$$

Which gives us

$$\langle 0 | \phi(\vec{x}, t) \phi(\vec{y}, t) | 0 \rangle = \langle 0 | \left[ \int \frac{d^3 k}{(2\pi)^3} \frac{d^3 q}{2E_q} e^{-ikx+iqy} \underbrace{[a_{\vec{k}}, a_{\vec{q}}^\dagger]}_{(2\pi)^3 \delta^{(3)}(\vec{k}-\vec{q})} \right] | 0 \rangle$$

$$\Rightarrow = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2E_k} e^{-ik(x-y)} \underbrace{\langle 0 | 0 \rangle}_{=1} \quad \text{Ground state (normalised to unity)}$$

$$\Rightarrow \langle 0 | \phi(\vec{x}, t) \phi(\vec{y}, t) | 0 \rangle = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2E_k} e^{-ik(x-y)}$$

$$\Rightarrow \langle 0 | \phi(\vec{y}, t) \phi(\vec{x}, t) | 0 \rangle = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2E_k} e^{-ik(y-x)}$$

The four-dimensional integral is written as

$$\int \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} \frac{i}{k^2 - \mu^2 + i\varepsilon} =$$

$$k_0^2 - (\vec{p}^2 + \mu^2) \rightarrow k_0 = \pm E_p$$

We have poles at  $k_0 = \pm (E_k - i\varepsilon)$

with this we get the integral into the form

since  $\varepsilon$  is infinitesimal  
 $\Rightarrow k_0^2 \approx E_p^2 - 2i\varepsilon \approx E_p^2 - i\varepsilon$   
 $\uparrow$  using Taylor expansion

$$\int \frac{d^3 \vec{k}}{(2\pi)^3} e^{-i\vec{k}(\vec{x}-\vec{y})} \underbrace{\int \frac{dk_0}{(2\pi)} \frac{i}{(k_0 - E_k + i\varepsilon)(k_0 + E_k - i\varepsilon)} e^{-ik_0(x_0 - y_0)}}_{\equiv I(k_0)}$$

To get the 3-dimensional integral we must solve  $I(k_0)$

$$I(k_0) = \int \frac{dk_0}{(2\pi)} \frac{+i}{(k_0 - E_k + i\varepsilon)(k_0 + E_k - i\varepsilon)} e^{-ik_0(x_0 - y_0)}$$

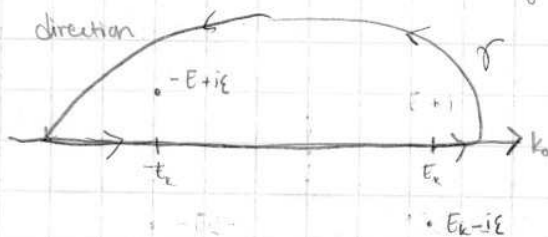
Let's first look at the case for  $x_0 > y_0$ :

We can solve the integral using complex analysis & Residue theorem according to which

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{i=1}^n \text{Res}(f, z_i)$$

where  $z_i$  are pole  $f(z)$  & the curve  $\gamma$  goes around the poles in the anticlockwise direction

in our case



the integral can be separated into two parts allowing us to solve  $I(k_0)$



From the definition of path integral we get

$$\oint_{\gamma} f(z) dz = \int_{\gamma(-R \rightarrow R)} f(z) dz + \int_{\gamma_R} f(z) dz$$

$\uparrow$   $\uparrow$   
 this is the integral this is the integral  
 along the number-line along the curve  
 from  $-R$  to  $R$

When we set  $R \rightarrow \infty$  the  $\int_{\gamma_R} f(z) dz$  term vanishes

So we get  $\oint_{\gamma} f(z) dz = \int_{-\infty}^{\infty} f(z) dz = 2\pi i \sum_{i=1}^n \text{Res}(f, z_i)$

in our case  $f(z) = +i \left[ (z-z_0)(z+z_0) \right]^{-1} e^{-iz(x-y)}$  with  $z_0 = E_k - i\varepsilon$   
 we only have one pole at  $z_0$  that is inside the curve  $\gamma$ .

The Residue is defined as

$$R(f, z_0) = \lim_{z \rightarrow z_0} [(z-z_0) f(z)] \quad \text{for simple pole } z_0$$

In our case we get

$$R(f, z_0) = \lim_{k_0 \rightarrow E_k - i\varepsilon} \left[ (k_0 - E_k + i\varepsilon) \frac{e^{-ik_0(x-y)}}{(k_0 - E_k + i\varepsilon)(k_0 + E_k - i\varepsilon)} \right]$$

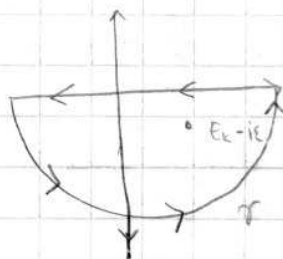
$$= +i \frac{e^{-i(E_k - i\varepsilon)(x-y)}}{2E_k - 2i\varepsilon} = +i \frac{e^{-iE_k(x-y)}}{2E_k} \quad \left( \begin{array}{l} \text{as } \varepsilon \rightarrow 0 \\ \text{we have } 2i\varepsilon \rightarrow 0 \end{array} \right)$$

$$\Rightarrow +i \int_{-\infty}^{\infty} \frac{dk_0 e^{-ik_0(x-y)}}{(k_0 - E_k + i\varepsilon)(k_0 + E_k - i\varepsilon)} = + \frac{2\pi i}{2E_k} e^{-iE_k(x-y)} \quad \leftarrow (E_k = k_0 \text{ so we substitute it back})$$

$$\Rightarrow I(k_0)_{x=y_0} = \frac{e^{-ik_0(x-y)}}{2E_k} \quad \text{for the case}$$

In the case  $y_0 > x_0$  the integral converges only in the bottom plane

We make a contour



$$\Rightarrow \oint_{\gamma} f(z) dz = \int_{-\infty}^{\infty} f(z) dz + \text{term that vanishes} = 2\pi i R(f, -z_0) = 2\pi i \lim_{z \rightarrow -z_0} [(z+z_0) f(z)]$$

$$\Rightarrow - \int_{-\infty}^{\infty} f(z) dz = \lim_{z \rightarrow -z_0} ((z+z_0) f(z)) \quad (z_0 = E_k - i\varepsilon) \text{ - which is pole in previous case}$$

$$\Rightarrow \int_{-\infty}^{\infty} f(z) dz = -2\pi i \lim_{k \rightarrow -E_k + i\epsilon} (z + E_k - i\epsilon) \frac{i}{[z - z_0][z + z_0]} e^{-ik(x_0 - y_0)}$$

$$i \int_{-\infty}^{\infty} \frac{dk e^{-ik(x_0 - y_0)}}{[z - z_0][z + z_0]} = -2\pi i \lim_{k \rightarrow -E_k + i\epsilon} \frac{e^{-ik(x_0 - y_0)}}{k - E_k + i\epsilon} = \frac{-2\pi i e^{-ik(x_0 - y_0)}}{-2E_k + 2i\epsilon} \stackrel{\epsilon \rightarrow 0}{\underset{k \rightarrow 0}{=}} \frac{2\pi i}{2E_k} e^{-E_k(y_0 - x_0)}$$

$$\Rightarrow I(k_0)_{y_0 > x_0} = \frac{1}{2E_k} e^{-ik(y_0 - x_0)} \quad \leftarrow (E_k = k_0 \text{ so we substitute } k_0 \text{ back in})$$

$$\text{So we get } I(k_0)_{\text{all}} = I(k_0)_{x_0 > y_0} + I(k_0)_{y_0 > x_0}$$

Writing more elegantly with Heaviside step function we get

$$I(k) = \theta(x_0 - y_0) I(k)_{x_0 > y_0} + \theta(y_0 - x_0) I(k)_{y_0 > x_0}$$

$$I(k) = \theta(x_0 - y_0) \frac{1}{2E_k} e^{-ik(x_0 - y_0)} + \theta(y_0 - x_0) \frac{1}{2E_k} e^{-ik(y_0 - x_0)}$$

$$\Rightarrow \int \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} \frac{i}{k^2 - \mu^2 + i\epsilon} = \int \frac{d^3 \vec{k}}{(2\pi)^3} e^{-i\vec{k}(\vec{x} - \vec{y})} I(k)$$

$$= \int \frac{d^3 \vec{k}}{(2\pi)^3} e^{-i\vec{k}(\vec{x} - \vec{y})} \frac{1}{2E_k} \left[ \theta(x_0 - y_0) e^{-ik(x_0 - y_0)} + \theta(y_0 - x_0) e^{-ik(y_0 - x_0)} \right]$$

$$\text{We note that } \int \frac{d^3 \vec{k}}{(2\pi)^3} e^{-i\vec{k}(\vec{x} - \vec{y})} \cdot \frac{1}{2E_k} = \langle 0 | \phi(\vec{x}, t) \phi(\vec{y}, t) | 0 \rangle$$

So we get

$$\int \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} \frac{i}{k^2 - \mu^2 + i\epsilon} = \theta(x_0 - y_0) \langle 0 | \phi(x) \phi(y) | 0 \rangle + \theta(y_0 - x_0) \langle 0 | \phi(y) \phi(x) | 0 \rangle$$

$$\Rightarrow \Delta_F(x-y) = \int \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} \frac{i}{k^2 - \mu^2 + i\epsilon}$$

## Exercise 2.9

Consider a complex valued scalar field with Lagrangian

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi$$

2) Conjugate momenta for  $\phi(x)$  is  $\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)}$

and for  $\phi^*(x)$  it is  $\pi^*(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^*(x)}$

$$\Rightarrow \pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)} = \dot{\phi}^*(x)$$

$$\pi^*(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^*(x)} = \dot{\phi}(x)$$

$\Leftarrow$  Conjugate momentum for each dynamical variable

$\Rightarrow$  The Hamilton then becomes

$$H = \int d^3x \left( \underbrace{\pi(x)}_{\pi^*(x)} \dot{\phi}(x) + \underbrace{\pi^*(x)}_{\dot{\phi}(x)} \dot{\phi}^*(x) - \mathcal{L} \right)$$

This additional variable comes from having 2nd dynamical variable  $\phi^*(x)$  in addition to first one ( $\phi(x)$ )

$$\Rightarrow H = \int d^3x \left[ \pi \pi^* + \pi^* \pi - \underbrace{(\dot{\phi}^* \dot{\phi})}_{= \pi \pi^*} - \nabla \phi^* \cdot \nabla \phi - m^2 \phi^* \phi \right]$$

$$H = \int d^3x \left( \pi^* \pi + \nabla \phi^* \cdot \nabla \phi + m^2 \phi^* \phi \right)$$

Canonical commutation relation

$$\left[ \underbrace{\hat{\phi}(x)}_{= \phi(x)}, \underbrace{\hat{\pi}(y)}_{= \pi(y) = \dot{\phi}^\dagger(y)} \right] = i \delta^{(3)}(x-y)$$

(assuming  $x_0 = y_0$   
for the exercise  
so that the  
comm. relations stay  
same)

$$\Rightarrow [\phi^\dagger(x), \pi^\dagger(y)] = i \delta^{(3)}(x-y), \quad [\phi(x), \phi(y)] = 0 = [\pi(x), \pi(y)]$$

And since  $\phi^\dagger$  &  $\phi$  are independent variables we get the relations:

$$[\phi(x), \phi^\dagger(y)] = 0 = [\pi(x), \pi^\dagger(y)]$$



The Heisenberg equation of motion is  $i \frac{\partial}{\partial t} \phi = [\phi, H]$

$$\Rightarrow i \frac{\partial}{\partial t} \pi_y^\dagger = [\pi_y^\dagger, H] \quad \parallel \frac{\partial}{\partial t} \pi_y^\dagger = \frac{\partial}{\partial t} \dot{\phi}_y = \frac{\partial^2}{\partial x^2} \phi_y \quad \parallel \phi_y \equiv \phi(y)$$

$$i \partial_t^2 \phi_y = [\pi_y^\dagger, H] \quad \parallel H = \int d^3x (\pi_x^\dagger \pi_x + \nabla \phi_x^\dagger \cdot \nabla \phi_x + m^2 \phi_x^\dagger \phi_x)$$

$$i \partial_t^2 \phi_y = \int d^3x \left[ \underbrace{[\pi_y^\dagger, \pi_x^\dagger \pi_x]}_{=0} + [\pi_y^\dagger, \nabla \phi_x^\dagger \nabla \phi_x] + m^2 \underbrace{[\pi_y^\dagger, \phi_x^\dagger \phi_x]}_{=-i \delta^{(3)}(x-y) \phi_x} \right]$$

Since  $[\pi_y^\dagger, \phi_x] = 0$  and  $[\pi_y^\dagger, \nabla_x] = 0$  we get

$$\begin{aligned} [\pi_y^\dagger, \nabla \phi_x^\dagger \nabla \phi_x] &= [\pi_y^\dagger, \nabla \phi_x^\dagger] \cdot \nabla \phi_x = \nabla \underbrace{[\pi_y^\dagger, \phi_x^\dagger]}_{=-i \delta^{(3)}(x-y)} \nabla \phi_x \\ &= [\pi_y^\dagger, \phi_x^\dagger]^\dagger \nabla^2 \phi_x \\ &= [\phi_x, \pi_y] \nabla^2 \phi_x = i \delta^{(3)}(x-y) \nabla^2 \phi_x \end{aligned}$$

Usefull relation for future  
 $\delta^{(3)\dagger}(x-y) = -\delta^{(3)}(x-y)$   
 which follows from  
 $[\pi_y^\dagger, \phi_x^\dagger]^\dagger = [\phi_x, \pi_y]$

We justify  $[\pi_y^\dagger, \nabla_x] = 0$  since it is equivalent to  $[\pi_y^\dagger, p_x]$  times some constant  
 and  $[\pi_y^\dagger, p_x] = 0$ .

$$\Rightarrow i \partial_t^2 \phi_y = \int d^3x \left[ i \delta^{(3)}(x-y) \nabla^2 \phi_x - i \delta^{(3)}(x-y) m^2 \phi_x \right]$$

$$i \partial_t^2 \phi_y = i (\nabla^2 - m^2) \phi_y \Rightarrow \underbrace{(\partial_t^2 - \nabla^2 + m^2)}_{=\partial_\mu \partial^\mu} \phi_y = 0$$

$$\boxed{0 = (\partial_\mu \partial^\mu + m^2) \phi} \quad \text{this is the Klein-Gordon equation for } \phi$$

$$i \frac{\partial}{\partial t} \pi_y = [\pi_y, H] = i \partial_t^2 \phi^\dagger = \int d^3x \left[ \underbrace{[\pi_y, \pi_x^\dagger \pi_x]}_{=0} + [\pi_y, \nabla \phi_x^\dagger \nabla \phi_x] + m^2 \underbrace{[\pi_y, \phi_x^\dagger \phi_x]}_{=-i \delta^{(3)}(x-y) \phi_x^\dagger} \right]$$

$$[\pi_y, \nabla \phi_x^\dagger \nabla \phi_x] = \nabla_x \phi_x^\dagger \nabla_x [\pi_y, \phi_x] = \nabla_x^2 \phi_x [\pi_y, \phi_x] = \nabla_x^2 [\pi_y, \phi_x] \phi_x^\dagger$$

$$= -i \nabla_x^2 \delta^{(3)}(x-y) \phi_x^\dagger = -i \delta^{(3)\dagger}(x-y) \nabla^2 \phi_x^\dagger$$

$$= i \delta^{(3)}(x-y) \nabla^2 \phi_x^\dagger$$

We now use the  
 relation  $\delta^{(3)\dagger}(x-y) = -\delta^{(3)}(x-y)$   
 that we derived earlier

$$\Rightarrow i \partial_t^2 \phi_y^\dagger = \int d^3x i \delta^{(3)}(x-y) \nabla^2 \phi_x^\dagger - i \delta^{(3)}(x-y) m^2 \phi_x^\dagger$$

$$i \partial_t^2 \phi_y^\dagger = i \nabla^2 \phi_y^\dagger - i m^2 \phi_y^\dagger$$

$$0 = (\partial_t^2 - \nabla^2 + m^2) \phi^\dagger \quad \parallel \partial_t^2 - \nabla^2 = \partial_\mu \partial^\mu$$

$$0 = (\partial_\mu \partial^\mu + m^2) \phi^\dagger$$

We have shown the Heisenberg equation of motion for  $\phi(x)$  is the Klein-Gordon equation

b)  $\phi$  &  $\phi^\dagger$  are independent variables so we must not have  $\phi = \phi^\dagger$

$$\text{Let's first try } \phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [a_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x}]$$

$$\& \phi^\dagger(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [b_p e^{-ip \cdot x} + b_p^\dagger e^{ip \cdot x}] \quad (b_p \neq a_p)$$

if we now try get  $\phi^\dagger$  from original  $\phi(x)$  we get

$$\phi^\dagger = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [a_p^\dagger e^{+ip \cdot x} + a_p e^{-ip \cdot x}] = \phi(x) \quad \text{so the form of}$$

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [a_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x}] \quad \text{doesn't work for complex field}$$

And we must have distinct creation and annihilation operators

We now try

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [a_p e^{-ip \cdot x} + b_p^\dagger e^{ip \cdot x}] \quad \text{which means that}$$

$$\phi^\dagger(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [b_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x}] \neq \phi(x)$$

$\Rightarrow$  so the field operators contain two distinct creation & annihilation operators. To construct diagonal Hamiltonian we also need to know  $\Pi(x)$ .

$$\Pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}^\dagger(x) = \frac{\partial}{\partial t} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [b_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x}]$$

$$p_0 = E_p = E_p = \omega_p$$

$$\Pi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [-iE_p b_p e^{-ip \cdot x} + iE_p a_p^\dagger e^{ip \cdot x}]$$



$$\pi(x) = \int \frac{d^3p}{(2\pi)^3} (-i) \sqrt{\frac{E_p}{2}} \left[ b_p e^{-ip \cdot x} - a_p^\dagger e^{ip \cdot x} \right] \Rightarrow \pi^\dagger = \int \frac{d^3p}{(2\pi)^3} \frac{(-i)}{\sqrt{2}} \sqrt{E_p} \left[ a_p e^{-ip \cdot x} - b_p^\dagger e^{ip \cdot x} \right]$$

Inserting the fields into equation for H, we get

$$H = \int d^3x \left[ \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{-i^2 E_p E_q}{2} [a_p e^{-ip \cdot x} - b_p^\dagger e^{ip \cdot x}] [b_q e^{-iq \cdot x} - a_q^\dagger e^{iq \cdot x}] \right. \\ \left. + \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{1}{2\sqrt{E_p E_q}} \nabla \cdot (b_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x}) \cdot \nabla (a_q e^{-iq \cdot x} + b_q^\dagger e^{iq \cdot x}) \right. \\ \left. + m^2 \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{2}{\sqrt{E_p E_q}} (b_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x}) \cdot (a_q e^{-iq \cdot x} + b_q^\dagger e^{iq \cdot x}) \right]$$

$$\nabla \cdot p^\dagger = \nabla \cdot (b_p e^{-ip \cdot x} + a_p^\dagger e^{ip \cdot x}) = (-i \vec{p} \cdot) (b_p e^{-ip \cdot x} - a_p^\dagger e^{ip \cdot x}) = \vec{p} \cdot \pi(x)$$

$$\Rightarrow H = \int d^3x \int \frac{d^3p d^3q}{(2\pi)^6 2} \left[ i^2 E_p E_q (a_p b_q e^{-i(p+q) \cdot x} - a_p a_q^\dagger e^{-i(p-q) \cdot x} - b_p^\dagger b_q e^{i(p-q) \cdot x} + b_p^\dagger a_q^\dagger e^{i(p+q) \cdot x}) \right. \\ \left. + \frac{1}{\sqrt{E_p E_q}} (-i \vec{p} \cdot) (-i \vec{q} \cdot) (b_p a_q e^{-i(p+q) \cdot x} - b_p b_q^\dagger e^{-i(p-q) \cdot x} - a_p^\dagger a_q e^{i(p-q) \cdot x} + a_p^\dagger b_q^\dagger e^{i(p+q) \cdot x}) \right. \\ \left. + \frac{m^2}{\sqrt{E_p E_q}} (b_p a_q e^{-i(p+q) \cdot x} + b_p b_q^\dagger e^{-i(p-q) \cdot x} + a_p^\dagger a_q e^{i(p-q) \cdot x} + a_p^\dagger b_q^\dagger e^{i(p+q) \cdot x}) \right]$$

$$H = \int d^3x \int \frac{d^3p d^3q}{(2\pi)^6 2} \left[ e^{-i(p+q) \cdot x} \left( i^2 \sqrt{E_p E_q} a_p b_q + \frac{i^2 \vec{p} \cdot \vec{q} + m^2}{\sqrt{E_p E_q}} b_p a_q \right) \right. \\ \left. + e^{-i(p-q) \cdot x} \left( \sqrt{E_p E_q} a_p a_q^\dagger + \frac{\vec{p} \cdot \vec{q} + m^2}{\sqrt{E_p E_q}} b_p b_q^\dagger \right) \right. \\ \left. + e^{i(p-q) \cdot x} \left( \sqrt{E_p E_q} b_p^\dagger b_q + \frac{\vec{p} \cdot \vec{q} + m^2}{\sqrt{E_p E_q}} a_p^\dagger a_q \right) \right. \\ \left. + e^{i(p+q) \cdot x} \left( i^2 \sqrt{E_p E_q} b_p^\dagger a_q^\dagger + \frac{i^2 \vec{p} \cdot \vec{q} + m^2}{\sqrt{E_p E_q}} a_p^\dagger b_q^\dagger \right) \right]$$

Now we perform the integral over  $d^3x$

$$\text{Using the identity } \delta(x-\alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ip(x-\alpha)} dp$$

we get the relations

$$\int d^3x e^{-i(p+q)x} = e^{-i(E_p+E_q)t} \underbrace{\int d^3x e^{+i(\vec{p}+\vec{q})\vec{x}}}_{=(2\pi)^3 \delta^{(3)}(\vec{p}+\vec{q})} = (2\pi)^3 e^{-i(E_p+E_q)t} \delta^{(3)}(\vec{p}+\vec{q})$$

$$\int d^3x e^{-i(p-q)x} = e^{-i(E_p-E_q)t} \int d^3x e^{+i(\vec{p}-\vec{q})\vec{x}} = (2\pi)^3 e^{-i(E_p-E_q)t} \delta^{(3)}(\vec{p}-\vec{q})$$

$$\int d^3x e^{i(p+q)x} = e^{i(E_p+E_q)t} \int d^3x e^{i(-\vec{p}-\vec{q})\vec{x}} = (2\pi)^3 e^{+i(E_p+E_q)t} \delta^{(3)}(\vec{p}+\vec{q}) \quad \left\| \delta(-x) = \delta(x) \right.$$

Similarly

$$\int d^3x e^{i(p-q)x} = e^{i(E_p-E_q)t} (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q})$$

Using these results we get

$$H = \int \frac{d^3p d^3q}{(2\pi)^6 2} \left[ (2\pi)^3 \delta^{(3)}(\vec{p}+\vec{q}) e^{-i(E_p+E_q)t} \left( i^2 \sqrt{E_p E_q} a_p b_q + \frac{m^2 - \vec{p} \cdot \vec{q}}{\sqrt{E_p E_q}} b_q a_q \right) \right. \\ + (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q}) e^{-i(E_p-E_q)t} \left( \sqrt{E_p E_q} a_p a_q^\dagger + \frac{m^2 + \vec{p} \cdot \vec{q}}{\sqrt{E_p E_q}} b_p b_q^\dagger \right) \\ + (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q}) e^{+i(E_p+E_q)t} \left( \sqrt{E_p E_q} b_p^\dagger b_q + \frac{m^2 + \vec{p} \cdot \vec{q}}{\sqrt{E_p E_q}} a_p^\dagger a_q \right) \\ \left. + (2\pi)^3 \delta^{(3)}(\vec{p}+\vec{q}) e^{+i(E_p+E_q)t} \left( i^2 \sqrt{E_p E_q} b_p^\dagger a_q^\dagger + \frac{m^2 - \vec{p} \cdot \vec{q}}{\sqrt{E_p E_q}} a_p^\dagger b_q^\dagger \right) \right]$$

Now we integrate over  $d^3q$  to get this term since its multiplier vanishes doesn't matter this  $e^k$  term comes from  $a_p = a_p \cdot e^k$  term

$$H = \int \frac{d^3p}{(2\pi)^3} \cdot \frac{1}{2} \left[ e^{-2iE_p t} \underbrace{\left( -E_p a_p b_p + \frac{m^2 + \vec{p}^2}{E_p} b_p a_p \right)}_{=0} e^k + (E_p a_p a_p^\dagger + E_p b_p b_p^\dagger) + (E_p b_p^\dagger b_q + E_p a_p^\dagger a_q) \right. \\ \left. + e^{2iE_p t} \underbrace{\left( -E_p b_p^\dagger a_p^\dagger + E_p a_p^\dagger b_p^\dagger \right)}_{=0} e^{k'} \right]$$

$\delta$ -function  
 here  $m^2 - \vec{p} \cdot \vec{q} \rightarrow m^2 - \vec{p} \cdot (-\vec{p})$   
 $= m^2 + \vec{p}^2 = E_p^2$   
 $\& [a_p, b_p] = 0$  so this term vanishes  
 $[a_p^\dagger, b_p^\dagger] = 0$

$$H = \int \frac{d^3p}{(2\pi)^3} \cdot \frac{1}{2} E_p [a_p a_p^\dagger + a_p^\dagger a_p + b_p^\dagger b_p + b_p b_p^\dagger]$$

Same here as in part before. Adding these after I completed the exercise & noticed something was missing so sorry for cluttered text.

Now we can either expand this in terms of commutators to get  $H$  with infinite vacuum energy which we have to ignore or just do normal ordering. We choose to do normal ordering since it is simpler.

$$\Rightarrow :H: = \int \frac{d^3p}{(2\pi)^3} \cdot \frac{1}{2} E_p [2a_p^\dagger a_p + 2b_p^\dagger b_p]$$

$$:H: = \int \frac{d^3p}{(2\pi)^3} E_p [a_p^\dagger a_p + b_p^\dagger b_p]$$

$\Rightarrow$  So our Hamiltonian has two sets of particles of mass  $m$  since both particles have same energy  $E_p$  when they have same momentum  $p$  since  $E_p = \sqrt{p^2 + m^2}$

$\Rightarrow$  The theory contains two sets of particles of mass  $m$ .

Exercise 2.10

The field is invariant under  $\phi \rightarrow e^{i\theta} \phi$ ,

$$\phi^\dagger \rightarrow e^{-i\theta} \phi^\dagger$$

We can start by looking at infinitesimal transformation such that  $\theta \ll 1$

$$\approx \phi \rightarrow \phi + i\theta \phi \quad \& \quad \phi^\dagger \rightarrow \phi^\dagger - i\theta \phi^\dagger$$

$\underbrace{\quad}_{=\delta\phi} \qquad \qquad \underbrace{\quad}_{=\delta\phi^\dagger}$

The Lagrangian transforms same as the field itself, so if only one of the field changes we get:

$$\Rightarrow L_\phi \rightarrow L(1 + i\theta) = L + i\theta L, \quad L_{\phi^\dagger} \rightarrow L(1 - i\theta) = L - i\theta L$$

$\delta L = \delta L_\phi = \theta \partial_\mu K_\phi^\mu = \theta \partial_\mu K^\mu$ 
 $\delta L_{\phi^\dagger} = -\delta L_\phi = -\theta \partial_\mu K_\phi^\mu = \theta \partial_\mu K_{\phi^\dagger}^\mu = -\theta \partial_\mu K^\mu$

$\uparrow$  Change only from  $\phi$ -change Contribution       $\uparrow$  Change only from  $\phi^\dagger$ -change Contribution

So the total change is  $L \rightarrow L + \delta L - \delta L = L$

$\Rightarrow \delta L = 0$  when we change  $\phi$  by  $i\theta\phi$  &  $\phi^\dagger$  by  $-i\theta\phi^\dagger$

From this & Noether's theorem we get the conserved current for field  $\phi$  &  $\phi^\dagger$  operators

$$j_\phi^\mu = \frac{\partial L}{\partial(\partial_\mu \phi)} i\phi - K^\mu \quad \& \quad j_{\phi^\dagger}^\mu = -\frac{\partial L}{\partial(\partial_\mu \phi^\dagger)} i\phi^\dagger + K^\mu$$

So for the whole field the total current is

$$j^\mu = j_\phi^\mu + j_{\phi^\dagger}^\mu = i \frac{\partial L}{\partial(\partial_\mu \phi)} \phi - i \frac{\partial L}{\partial(\partial_\mu \phi^\dagger)} \phi^\dagger$$



The conserved charge is then

$$Q = \int d^3x j^0 = i \int d^3x \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} \phi - \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi^\dagger)} \phi^\dagger$$

$$Q = i \int d^3x \dot{\phi}^\dagger \phi - \dot{\phi} \phi^\dagger$$

// we note from previous exercise  
that  $\Pi = \dot{\phi}^\dagger$  &  $\Pi^\dagger = \dot{\phi}$

$$Q = i \int d^3x \Pi \phi - \Pi^\dagger \phi^\dagger$$

// we now write  $\Pi, \phi, \phi^\dagger, \Pi^\dagger$   
in terms of creation & annihilation  
operators

$$Q = i \int d^3x \left[ \frac{d^3p d^3q}{(2\pi)^6 2} \left[ \sqrt{\frac{E_p}{E_q}} (-i) (b_p e^{-ipx} - a_p^\dagger e^{ipx}) (a_q e^{-iqx} + b_q^\dagger e^{iqx}) \right. \right. \\ \left. \left. - \sqrt{\frac{E_q}{E_p}} (-i) (a_p e^{-ipx} - b_p^\dagger e^{ipx}) (b_q e^{-iqx} + a_q^\dagger e^{iqx}) \right] \right]$$

$$Q = -i^2 \int d^3x \left[ \frac{d^3p d^3q}{(2\pi)^6 2} \sqrt{\frac{E_p}{E_q}} \left[ e^{-i(p+q)x} (b_p a_q - a_p b_q) \right. \right. \\ \left. \left. + e^{-i(p-q)x} (b_p b_q^\dagger - a_p a_q^\dagger) + e^{+i(p-q)x} (-a_p^\dagger a_q + b_p^\dagger b_q) \right. \right. \\ \left. \left. + e^{+i(p+q)x} (-a_q^\dagger b_q^\dagger + b_p^\dagger a_q^\dagger) \right] \right]$$

We use the deltafunction identity  $\int_{-\infty}^{\infty} dx e^{+i(p-x)x} = 2\pi \delta(p-x)$

$$\Rightarrow \int d^3x e^{-i(p+q)x} = \int d^3x e^{-i(E_p+E_q)t} e^{+i(\vec{p}+\vec{q})\cdot\vec{x}} = (2\pi)^3 e^{-i(E_p+E_q)t} \delta^{(3)}(\vec{p}+\vec{q})$$

Similarity  $\int dt e^{-i(p-q)x} = (2\pi)^3 e^{-i(E_p-E_q)t} \delta^{(3)}(\vec{p}-\vec{q})$

$$\int dt e^{+i(p-q)x} = (2\pi)^3 e^{+i(E_p-E_q)t} \delta^{(3)}(\vec{p}-\vec{q})$$

$$\int dt e^{+i(p+q)x} = (2\pi)^3 e^{+i(E_p+E_q)t} \delta^{(3)}(\vec{p}+\vec{q})$$

Continuing with the conserved charge after doing the  $d^3x$  &  $d^3q$  integrals we get

$$Q = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} \left[ e^{-2iE_p t} \underbrace{(b_p a_p - a_p b_{-p})}_{=0} + (b_p b_p^\dagger - a_p a_p^\dagger) + (b_p^\dagger b_p - a_p^\dagger a_p) + e^{2iE_p t} \underbrace{(-a_p^\dagger b_p^\dagger + b_p^\dagger a_p^\dagger)}_{=0} e^{k'} \right]$$

$$\begin{aligned} b_{-p} &= b_p \cdot e^k \\ a_{-p} &= a_p \cdot e^k, \text{ we don't} \\ &\text{care what the } k \text{ is since} \\ (b_p a_p - a_p b_p) &= 0 \text{ since } a_p \text{ \& } b_p \text{ commute.} \\ b_p^\dagger &= b_p^\dagger \cdot e^{k'} \\ a_p^\dagger &= a_p^\dagger \cdot e^{k'} \end{aligned}$$

$$Q = \int \frac{d^3p}{(2\pi)^3} \cdot \frac{1}{2} (b_p b_p^\dagger + b_p^\dagger b_p - a_p a_p^\dagger - a_p^\dagger a_p)$$

To avoid the infinite valued terms we do the normal ordering & get:

$$:Q: = \int \frac{d^3p}{(2\pi)^3} (b_p^\dagger b_p - a_p^\dagger a_p)$$

The two distinct particles  $b_p^\dagger$  &  $a_p^\dagger$  have distinct signs in the conserved charge equation  $\Rightarrow$  These two distinct particles have equal but opposite charges.